Two-Sample Mean Inference

Testing a difference in means

Parameter of interest: $\mu_1 - \mu_2$

- Point estimate: $\bar{X}_1 - \bar{X}_2$
- Standard error: $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
- Model: T_w



New example: The lizard data records the spike length of live lizards and those killed by shrikes.

Our parameter of interest is $\mu_1 - \mu_2$, where μ_1 is the average spike length for killed lizards, and μ_2 is the average spike length for living lizards.

We have $\bar{X}_1 - \bar{X}_2$ as our point estimate, and

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

as our standard error.

Can we use a regular T test approach?

$$T = \frac{\text{pt. est.} - \text{null}}{\text{est. std. error}} = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\text{est. std. error}}$$

A complication: we have to estimate *both* σ_1^2 and σ_2^2, which might be different.

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \text{becomes} \quad \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

The statistic

$$T = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

is only approximately T distributed. And the df are messy!



The **Welch** degrees of freedom are given by

$$w = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(S_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(S_2^2/n_2\right)^2}{n_2 - 1}}$$

We are penalized if S_1 and S_2 are very different.

Let’s return to our lizard data and set up a Welch T test.

For this test to be valid, we need:

- Both groups are normal by CLT
- Each group made of up independent observations
- Groups are independent of *each other*

Do longer spikes protect against being killed by shrikes? Our research question suggests *one-sided* hypotheses:

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad \mu_1 - \mu_2 > 0$$

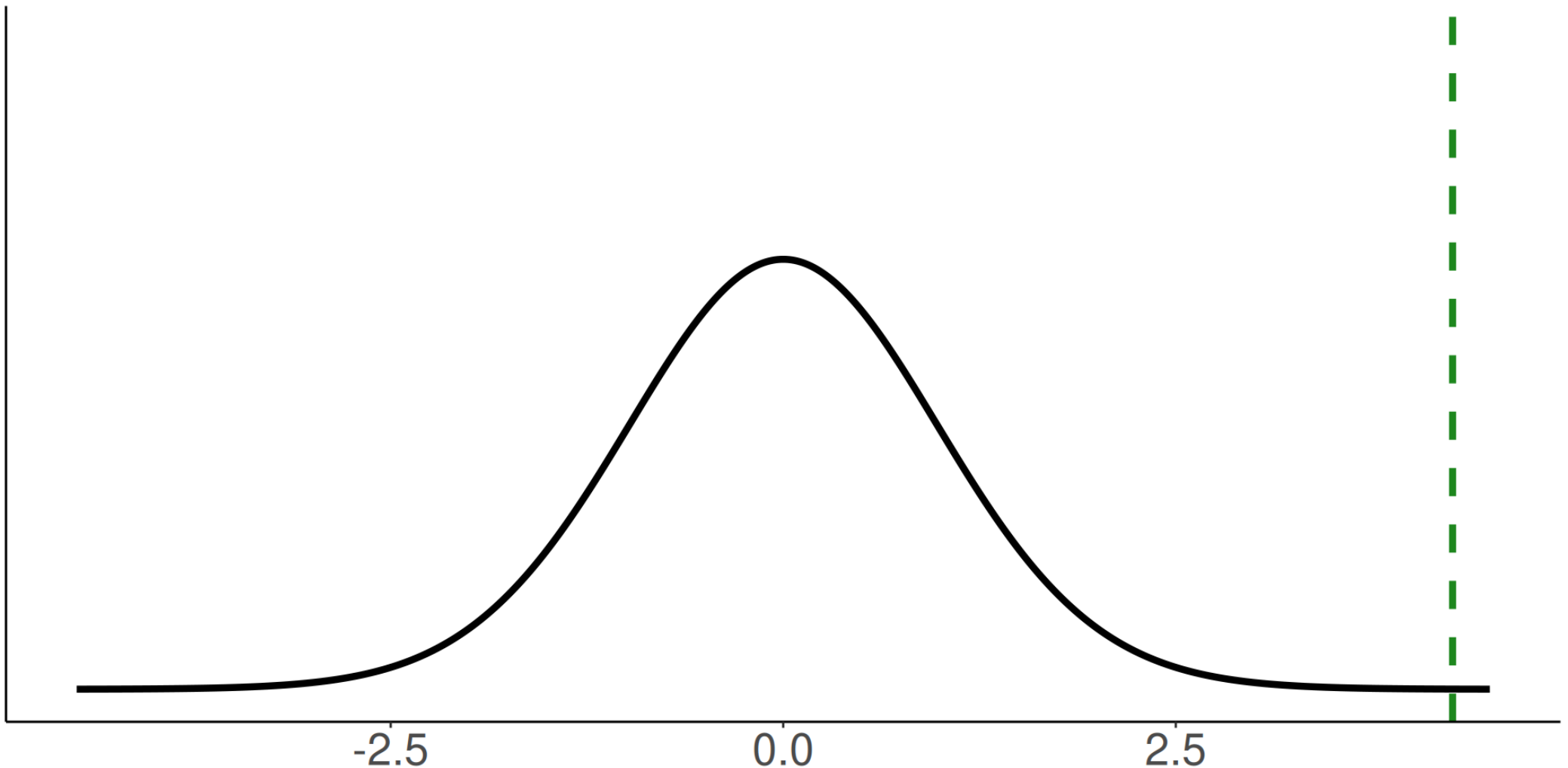
(Group 1 is alive, group 2 is dead). Let's complete this test with $\alpha = 0.01$.

Our observed test statistic is

$$t_{obs} = \frac{24.28 - 21.98 - 0}{\sqrt{\frac{2.63^2}{154} + \frac{2.71^2}{30}}} = 4.263$$

and the null T has about 40.37 df.

The one-sided p-value is the area *above* our test statistic.



We get a very small p-value of 0.00005, confirmed by `t.test()`.

We have strong evidence that the average spike length is *larger* for alive lizards compared to dead ones.



Exercises | Solutions/Hints

Continue the analysis of the lizard data with a 99% T CI of $\mu_1 - \mu_2$:

$$\text{point estimate} \pm \text{critical value} \times \text{standard error}$$

- Identify the three components of our interval.
- Interpret your interval in context.
- What happens if we reverse the order of subtraction?



$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 \neq 0$$

the Welch procedure uses test statistic

$$T = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

and has a T null distribution with w degrees of freedom. This works if we have independence and normality for both groups, and the groups themselves are independent.

If the conditions for the Welch test are met, we can also build a CI for $\mu_1 - \mu_2$:

$$\bar{X}_1 - \bar{X}_2 \pm t_{w,\alpha/2} \times \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

There is an exact two-sample T test, but it requires an additional assumption.

We need to assume the two groups have the same population variance (and therefore the same population SD)

$$\sigma_1^2 = \sigma_2^2 = \sigma^2 \qquad \sigma_1 = \sigma_2 = \sigma$$

The equal variances assumption is reasonable if the sample SDs are within a factor of two. We want:

$$0.5 < \frac{s_1}{s_2} < 2$$

This is true for the lizard data.

If we make the equal variances assumption, we can simplify the standard error of $\bar{X}_1 - \bar{X}_2$.

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \Rightarrow \quad \sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

We estimate σ, the common SD, by pooling the two sample SDs together.

$$s_p = \sqrt{\frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}}$$

s_p^2 is a *weighted* average of the two individual variances, where the weights are the individual degrees of freedom.

Then $\sqrt{s_p^2} = s_p$ estimates σ.

Exercises | Solutions/Hints

$$s_p = \sqrt{\frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}}$$

- Is s_p more similar to s_1 or s_2?
- When is s_p^2 an exact average of s_1^2 and s_w^2?
- CHALLENGE: Why do we square the SDs, then take an average, then square root? Why not just average s_1 and s_2?



Let's redo the lizard example with the equal variances assumption.

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 > 0$$

The pooled SD for the lizard data is about 2.64, which gives test statistic

$$t_{obs} = \frac{24.28 - 21.98 - 0}{2.64\sqrt{\frac{1}{154} + \frac{1}{30}}} = 4.35$$

which has a T null distribution with $n_1 + n_2 - 2 = 182$ df.

Again, we reject the null due to very strong evidence of a positive difference in the groups.

Within `t.test`, we can set `var.equal = T` to make the equal variance assumption.

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 \neq 0$$

the equal variances T procedure uses test statistic

$$T = \frac{\bar{X}_1 - \bar{X}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

and has a T null distribution with $n_1 + n_2 - 2$ df. This test is *less* flexible than the Welch test, because it has all the same assumptions plus an additional equal variance assumption.

The equal variances CI approach also uses a T with $n_1 + n_2 - 2$ df and the pooled SD.

$$\bar{X}_1 - \bar{X}_2 \ \pm \ t_{n_1+n_2-2,\alpha/2} \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Which test do we prefer?

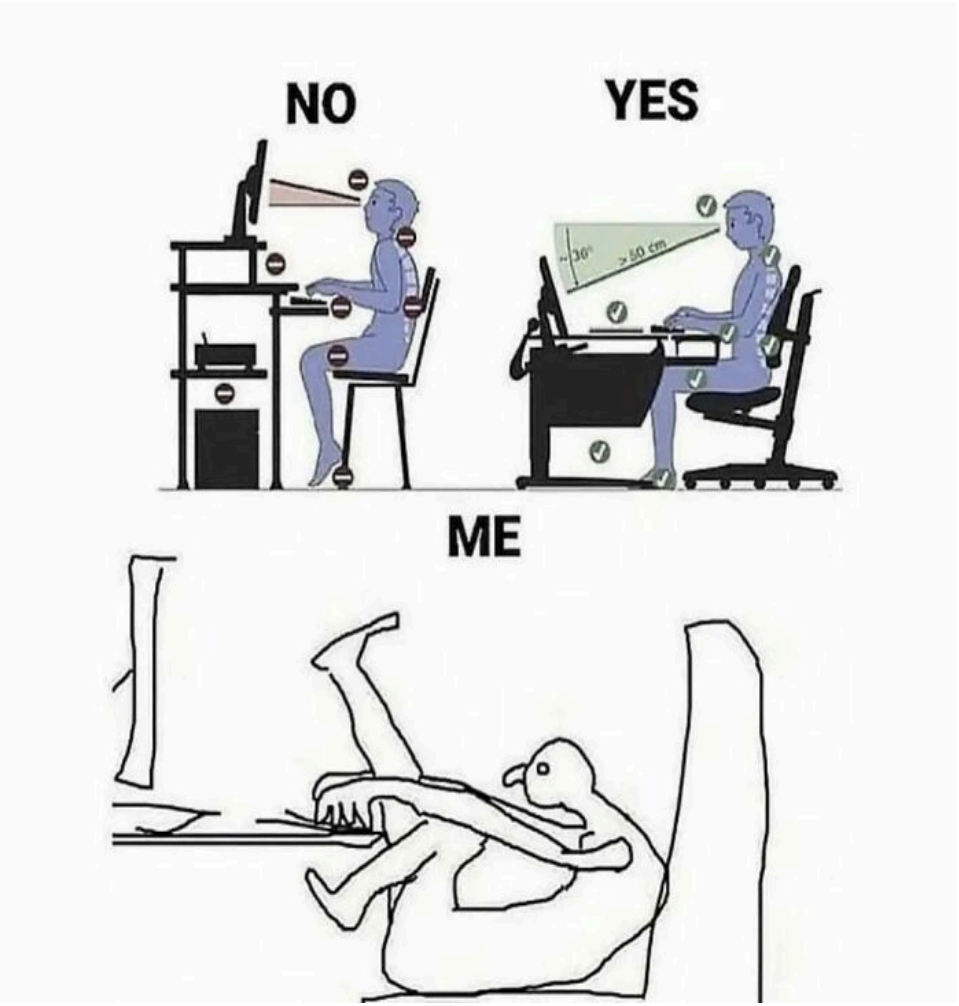
- If $\sigma_1 = \sigma_2$ but we use a Welch test, we lose a bit of statistical power.
- If $\sigma_1 \neq \sigma_2$ but we assume equal variances, we can make inaccurate conclusions.

The Welch procedure is generally preferred to equal variances. If the normality assumption is not met, both T test methods are unreliable and we should use simulation.

Sometimes, samples of data are *paired*.

Long periods of upper-arm elevation, possibly due to repetitive office work, can cause health issues.

Eight office workers are given standard work tasks one day, then varied work tasks another day.

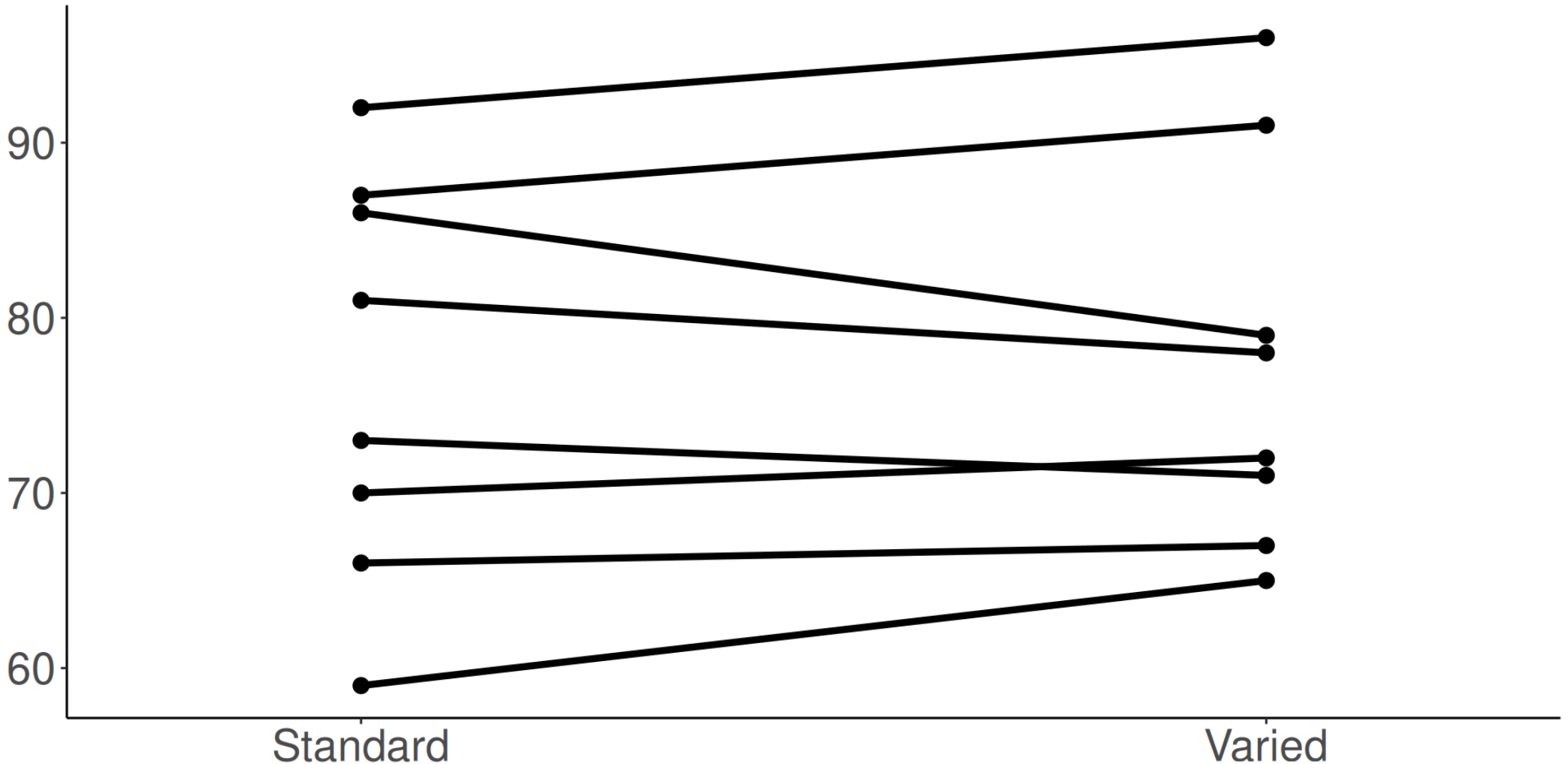


For each day, we record the percentage of time the worker's upper-arm angle was below 30 degrees.

Participant	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Standard	81	87	86	59	66	70	73	92
Varied	78	91	79	65	67	72	71	96

These two samples are *not* independent.

Paired observations aren’t independent, so the Welch or equal variances won’t work.



However, we can study the *differences*.

Participant	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Standard	81	87	86	59	66	70	73	92
Varied	78	91	79	65	67	72	71	96
Difference	3	-4	7	-6	-1	-2	2	4

The problem reduces to a one-sample T test!

$$H_0 : \mu_{diff} = 0 \quad \text{instead of} \quad H_0 : \mu_1 - \mu_2 = 0$$

Let's test

$$H_0 : \mu_{diff} = 0 \quad \text{versus} \quad H_A : \mu_{diff} \neq 0$$

at the 5% level.

The assumptions for a T test (independence, normality) apply to the differences.

Our test statistic is

$$T = \frac{\bar{x}_{diff} - \mu_0}{s/\sqrt{n}}$$

and the null distribution is a T with $n - 1$ df, where n is the number of *pairs*.

In the office worker example, we have a large p-value and a non-significant result.

Looking ahead

In chapter 14, we'll start learning how to fit a linear regression model between two numeric variables. Return to the `monona` data that records the freeze duration by year for Lake Monona. Suppose we use the model

$$y = 537.02 - 0.224x$$

where y is freeze duration and x is year. Using this model, what is the predicted freeze duration in year 1950? What was the actual freeze duration in that year? Did the model over- or under- estimate the duration, and by how much?